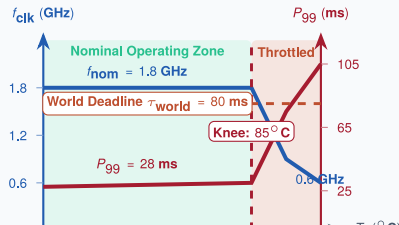


(A) THE THERMAL DERATING KNEE ($T_j > 85^{\circ}\text{C}$)

Dynamic Voltage & Frequency Scaling (DVFS) Inflates P_{gg} Latency Tail



Physical Systems Consequence:

When $T_j > 85^{\circ}\text{C}$, DVFS throttles clock from $1.8 \text{ GHz} \rightarrow 600 \text{ MHz}$. Inference latency triples ($P_{\text{gg}} = 105 \text{ ms} > \tau_{\text{world}}$), inflating physical stopping distance d_{stop} from $0.35 \text{ m} \rightarrow 0.82 \text{ m}$, breaching safety clearance.

Cloud vs. Edge NPU Placement Comparison:

Systems Dimension	Cloud 70B VLA	Edge 3B INT4 NPU
Model Parameter Scale	70 Billion FP16	3 Billion INT4
Inference Tail (P_{gg})	480 ms (High Jitter)	35 ms (Deterministic)

Engineering Rule of Thumb:

If $d_{\text{stop}}(\text{Cloud}) > d_{\text{clearance, min}}$, local Edge NPU placement is non-negotiable regardless of on-board battery and thermal costs.